



CERTIFIED MATERIAL TEST REPORT
CERTIFICATE OF COMPLIANCE
CERTIFICATE # 226593
REVISION 0

Tube Forgings of America, Inc.
5200 N.W. Front Avenue
Portland, OR 97210
ISO 9001:2015 Certified
4/9/2025

Sold To	DNOW LP #MI04 1701 Gover Pkwy. Mt Pleasant, MI 48858-8140	Ship To	DNOW LP #MI04 1701 Gover Pkwy. Mt Pleasant, MI 48858-8140
Customer ID	34217		
TFA Sales Order	326358\1-1		
Purchase Order	662682080		

TFA Part ID	00059	Heat Code	A2511
Description	4 XH LR 90	Grade	WPB
Quantity	74	Raw Material Heat	027614
Customer Part ID	3674	Process	WPB-H1
		Mill	TPCO

Austenitized @: Held For:	mins/min	Tempered @: Held For:	mins/min	Stress Relieved @: Held For:	mins/min	Medium:
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Heat Analysis Chemistry

C	Mn	P	S	Si	Mo	Cr	Ni	Cu	V	Cb	B	Ti	Al	N	Co	Ca	C.E.
0.15	1.06	0.010	0.002	0.26	0.01	0.06	0.02	0.02	0.01								0.35

Tensile Test Results

Test	Name	Yield	Tensile	% Elong	R/A %	Gage Length	HBW	In-Process Test 1	Test 2	Tensile Orientation	Test Bar Type & Size
1	BASE	45686 (psi)	70343 (psi)	36.00		2"	140	137	143	LONGITUDINAL	STRIP

Charpy Impact Test Results

	Impact Energy				Lateral Expansion (Mils)			% Shear				Test Temp.	Size (mm)	Charpy Orientation
	1	2	3	Avg	1	2	3	1	2	3	Avg			

Notes/Comments

1.THESE FITTINGS WERE HOT FORMED BETWEEN 1150FAND1800F AND AIR COOLED. 2.THE HARDNESS DOES NOT EXCEED 197 BHN PART TO PART. 3. STARTING MATERIAL IS SEAMLESS PIPE WHICH MEETS THE MELTING,CHEMICAL AND PHYSICAL REQUIREMENTS OF ASTM/ASME A/SA 106 GR B. 4. THIS PRODUCT IS SEAMLESS.

1: MATERIAL IS MANUFACTURED TO ASTM A234-19 GR WPB, ALSO PER ASME SEC II SA234 GR WPB-2019 EDITION. 2: DESIGN AND DIMENSIONS TO ASME B16.9-2018 3: PER NACE MR0175/ISO15156-2-2015 EDITION, AND NACE MR-0103 2015 EDITION. 4: MANUFACTURED IN AN ISO 9001 REGISTERED FACILITY, CERTIFICATE #: 30248. 5: TFA, INC. ORGANIZATIONAL STRUCTURE AND CERTIFICATION COMPLIES WITH EN 10204-2004 TYPE 3.1. 6: TFA INC. FITTINGS SATISFY THE MATERIAL REQUIREMENT OF (PED) 2014/68/EU ANNEX I , PARA 4.3 7: FITTINGS ARE MANUFACTURED IN THE USA. RAW MATERIAL IS FROM DOMESTIC AND GLOBAL SOURCES. 8: FITTINGS WERE MANUFACTURED, SAMPLED, TESTED AND INSPECTED IN ACCORDANCE WITH THE ORDER AND SPECIFICATIONS AND WERE FOUND TO MEET THE REQUIREMENTS.

*UB = Unbroken Test

Heat: A2511 Grade: WPB

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Scot Marshall, Manager Quality Assurance

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